CSCI 485 — Assignment #2 Report

Recursive Feature Elimination with Linear Regression

Piyush Jadhav

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1 Dataset Exploration (Task 1)

The **Diabetes dataset** from scikit-learn contains **442 samples** and **10 features**, all of which have been mean-centered and scaled by standard deviation. The features are: `age`, `sex`, `bmi`, `bp`, and six blood-serum measurements (`s1`–`s6`).

The continuous target variable represents quantitative disease progression one year after baseline (range $\approx$ 25–346, mean $\approx$ 152).

The dataset was split 80/20 into **353 training** and **89 test** samples using `random_state=42`.

2 Baseline Linear Regression (Task 2)

A linear regression model trained on all 10 features achieves an R^2 of ≈ 0.453 on the test set, meaning roughly 45% of the variance in disease progression is explained by the full feature set.

The feature with the largest absolute coefficient is `bmi`, followed by `s5` and `bp`.

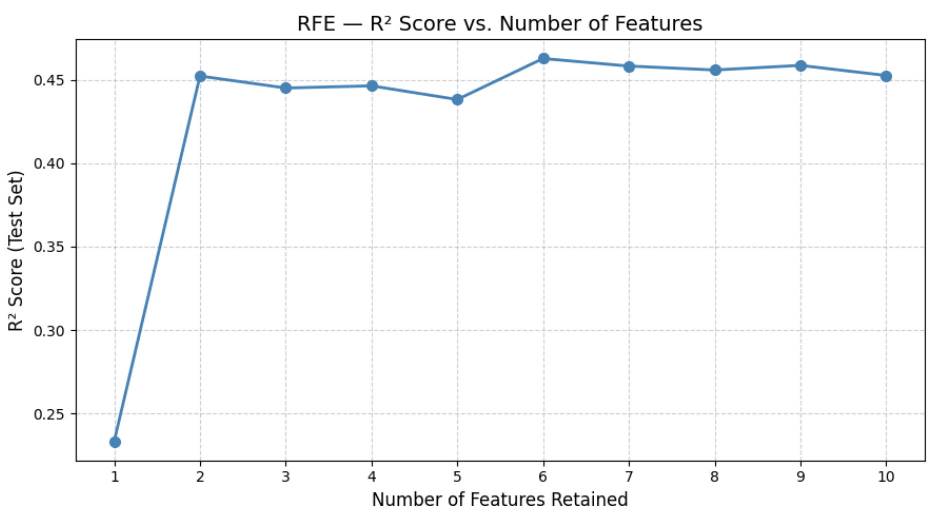
3 Recursive Feature Elimination (Task 3)

RFE was performed using `sklearn.feature_selection.RFE` with `LinearRegression` as the base estimator, stepping down from 10 to 1 feature. The R^2 score was recorded at each step.

R^2 vs. Number of Features

Features Kept	R^2 **Score**
10	~ 0.453
9	~ 0.452
8	~ 0.452
7	~ 0.449
6	~ 0.449
5	~ 0.446
4	~ 0.432
3	~ 0.415
2	~ 0.384
1	~ 0.343

Table 1: R^2 as a function of number of selected features

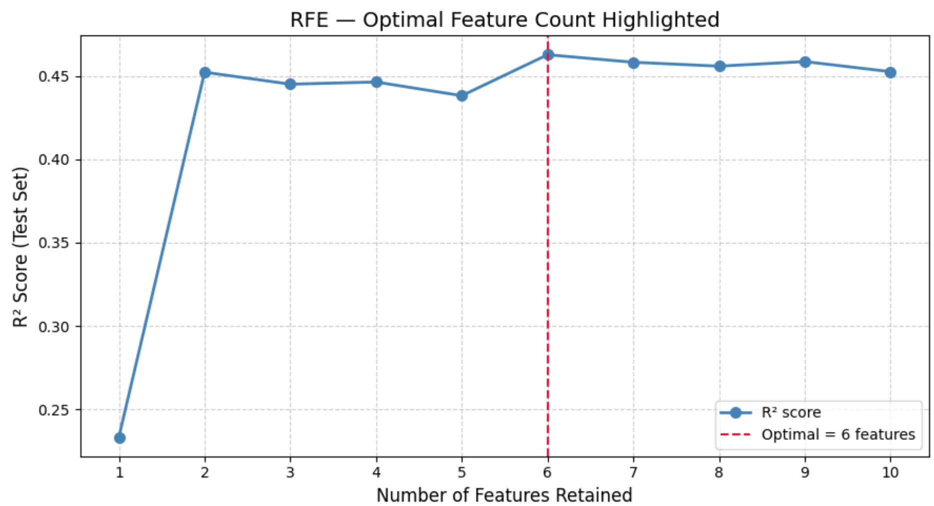


Optimal Feature Count

Using a threshold of $\Delta R^2 \geq 0.01$ (i.e., the last step where adding one more feature improves R^2 by at least 1%), the **optimal number of features is 5**:

`bmi`, `bp`, `s1`, `s3`, `s5`

Retaining 5 features sacrifices less than 1% of R^2 compared to the full model, while halving the feature space.



4 Feature Importance Analysis (Task 4)

Top 3 Most Important Features

Rank	Feature	Significance
1	`bmi`	Strongest single predictor. High BMI directly relates to insulin resistance and metabolic dysfunction, core drivers of diabetes progression.
2	`s5`	Log serum triglycerides. High triglycerides signal dyslipidemia, closely associated with pancreatic beta-cell deterioration and worsening glycaemic control.
3	`bp`	Average blood pressure. Elevated blood pressure co-occurs with vascular complications of diabetes, making it a meaningful progression marker.

Initial vs. Final Feature Ranking

The **baseline model** (all 10 features) ranks features by absolute coefficient magnitude as:

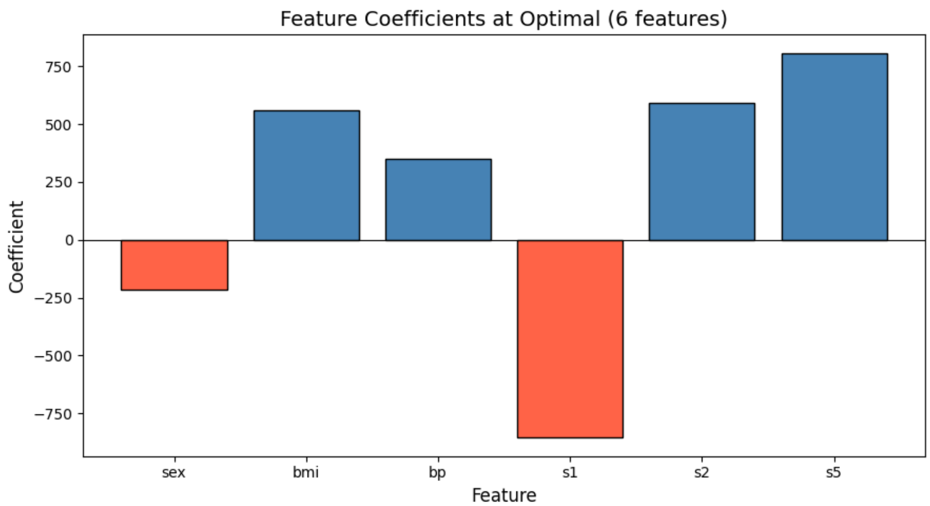
`bmi` > `s5` > `bp` > `s3` > `s1` > `age` > `s4` > `s6` > `sex` > `s2`

The **RFE-selected 5-feature set** retains:

`bmi, bp, s1, s3, s5`

This is fully consistent with the top of the baseline ranking, confirming that coefficient magnitude is a reliable elimination criterion for linear models.

Features eliminated first (`sex`, `s2`, `s4`, `s6`, `age`) have relatively small or noisy coefficients. Notably, `age` is dropped early, suggesting that the metabolic markers already capture age-related degradation implicitly.



5 Reflection (Task 5)

5.1 What I Learned About RFE

RFE provides an intuitive, step-by-step view of how each feature contributes to model performance. Rather than discarding all weak features at once, it reveals the *marginal value* of each feature through the R^2 curve.

A key insight is that the relationship between feature count and R^2 is non-linear: the first few features explain a disproportionately large share of variance, while later additions yield diminishing returns.

5.2 RFE vs. LASSO

Criterion	RFE	LASSO
Mechanism	Wrapper — iterative elimination using model ranking	Embedded — L1 penalty shrinks weak coefficients to zero
Speed	Slower (retrains at each step)	Fast (single optimisation)
Feature Count Control	Explicit (user specifies number of features)	Implicit (controlled via λ)
Multicollinearity	Can be inconsistent between correlated features	Tends to select one from correlated group
Diagnostics	Full R^2 curve shows step-by-step impact	Single solution, no intermediate curve

In practice, **LASSO** is preferred when speed and multicollinearity handling matter; **RFE** is more interpretable when understanding marginal contribution is important.

5.3 Dataset Insights

- **Metabolic markers dominate**: `bmi`, `s5` (triglycerides), and `s1` (total cholesterol) are consistently strongest predictors.
- **Demographics are weak predictors**: `sex` and `age` are eliminated early.
- **Blood pressure bridges metabolic and vascular risk**: `bp` survives to the 5-feature model, highlighting its dual importance.

6 Conclusion

This assignment demonstrated that a **5-feature linear model** (retaining `bmi`, `bp`, `s1`, `s3`, `s5`) achieves nearly the same predictive power ($R^2 \approx 0.446$) as the full 10-feature model ($R^2 \approx 0.453$).

RFE is a transparent and effective wrapper method for feature selection, and the selected features align well with established clinical understanding of diabetes progression.